

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · PHYSICS

PHYSICS 5054/2
PAPER 2 November 2024
2 hours

Candidate Name: _____

Centre Number: _____ **Candidate Number:** _____

TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES

Electronic calculators may be used. Take $g = 10 \text{ m/s}^2$ (or 9.8 m/s^2).

Show all formulas, substitutions, working, and correct units.

SECTION A

1. A car of mass 1200 kg accelerates uniformly from rest to 20 m/s in 8.0 seconds. (a) Calculate the acceleration of the car. (b) Determine the resultant forward force on the car. (c) Calculate the kinetic energy of the car at 20 m/s. *[Mechanics & Newton's Laws]* [6]
- (a) Calculate the acceleration. [2]
- (b) Determine the accelerating force. [2]
- (c) Calculate the kinetic energy. [2]